Tel Aviv University

The Iby and Aladar Fleischman Faculty of Engineering

Electronics - Laboratory (3)

Experiment FPGA 6 – VGA

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Based on Konstantin Berestizshevsky's work

Lab Goals (This is also the material for the test)

The purpose of this lab is to cover the following topics:

1. Video Graphics Array (VGA) interface – synchronization signals, color space encoding, graphics representation.

Submission Guidelines

The lab consists of TBD tasks, where each task depends on the previous.

1. Submit a report (PDF format), as well as the XDC constraints file, and all the *.v files you have edited. All zipped together in a zip file named EE3_FPGA_3 <ID1> <ID2>
2. Each waveform must be annotated by (i) drawing arrows that point to important signals and (ii) explaining these signals.
3. Most of the modules in this lab are parametric (for example, Compadder(N) has a design parameter N, which determines the bit-width of its operands). These modules must in fact support different parameter values, and not to stick to some constant value.
4. Each Verilog source/test-bench header must have both of your names in it:

```
`timescale 1ns/10ps
//////////////////////////////////////////////
// Company:          Tel Aviv University
// Engineer:         Your name
```

5. At home you can work with the Vivado Web-Pack. You will need to create a free student account in Xilinx before downloading and installing the software.
6. It is your responsibility to either use a USB-flash drive or other cloud solutions to save your project before you leave the laboratory. The laboratory PCs are erased from time to time.

1. Vivado project

Launch the Vivado software, create a new project. Follow the “Vivado Project Creation” subsection of the Vivado-tutorial to set up a new project. The name of the project should be **lab6**, and it should be saved in **G:\FPGA_LAB** path. The project type is **RTL Project**. Make sure to choose the **xc7a35tlcpg236-2L** part in the “Default Part” screen of the wizard. Notes:

- To add more sources later, go to “Project Manager→Add Sources” in the flow-navigator in the left part of the Vivado screen.
- In this lab, all the tasks are to be performed under the same “lab6” project.

Task 2 – VGA interface for screen output

In this part of the lab you will design a VGA interface to output graphics to the computer monitor connected to the Basys3 board. So far, we were limited to either the 7-segment display or the LEDs. In this lab, we expand this functionality to allow graphical images to be displayed from the FPGA board. And we will examine the functionality of your design by displaying a rectangle and control its size and color. Figure depicts a high-level diagram of a design required in this task.

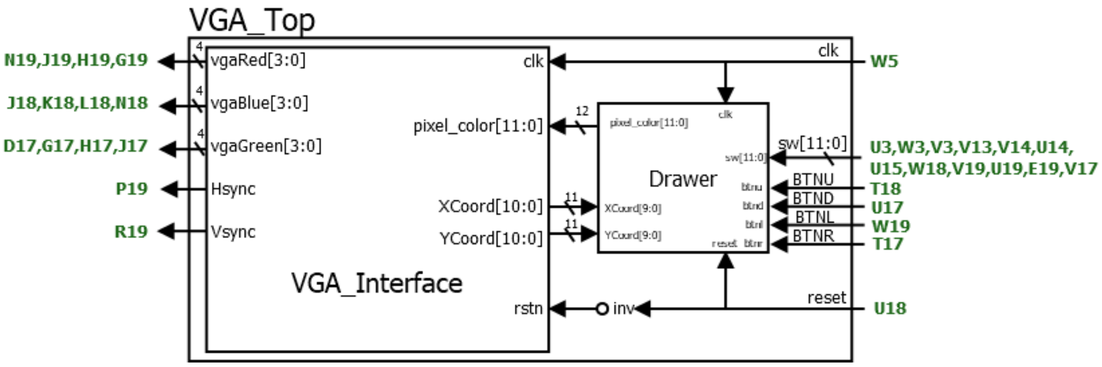

Figure 1 - High level diagram of the design required in the task. The green labels are the FPGA pin name whereas the black labels are the IO ports of the Verilog sources.

The modules that you are required to implement are:

1) **VGA_interface** – with 3 inputs: clk, rstn, pixel_color; and 7 outputs: vgaRed, vgaGreen, vgaBlue, Hsync, Vsync, XCoord, YCoord. The VGA interface takes as an input a 12-bit signal pixel_color (which contains 4-red-channel bits, 4-green-channel bits, 4 blue-channel bits) which represents a color of a single pixel, and outputs this color to the VGA-output ports. The VGA interface will generate 50MHz pixel-clock internally. The outputs vga* and *sync of VGA interface must respect the VGA standard for 800x600 resolution (see theoretical background). It will also output the current horizontal coordinate (0-1039) and vertical coordinate (0-665).
2) **Drawer** – 9 inputs: clk, reset, sw (12 bits), btnu(1 bit), btnd(1 bit), btnl(1 bit), btnr(1 bit), XCoord (11 bits), YCoord (11 bits). The output is the 12-bit encoding of the current color to be shown on the screen. This is a module which decides which color is to be sent to the current pixel on the screen, where (sw[11:8], sw[7:4], sw[3:0]) represents the (R,G,B) channels. At start, the rectangle shall appear in the center of the screen with size 1 (i.e. a dot), pressing the btnu, btnd, btnl and btnr shall expand the rectangle to the corresponding direction.
3) **VGA_Top** – a top level module comprised of the VGA_Interface and the Drawer modules. It receives clock and reset signals, as well as the relecant buttons and switches from the board. It outputs the VGA control signals using VGA_Interface.

Guidance: The VGA interface can be implemented as follows:

1. Warning: the 72Hz screen refresh rate may not be supported by some of the monitors!
2. Use 2 counters to store the values of *hcount* and *vcount*
3. On the rising edge of the pixel clock, increment the *hcount*. Increment *vcount* when *hcount* has reached the end of the row.
4. Generate the *Hsync* signal based on the value of *Hcount*. *Vsync* is generated in a similar fashion. Refer to the theoretical background for timing diagrams.
5. Generate a signal to determine whether the pixel is in the visible region as illustrated in the regions diagram in theoretical background

6. When in the visible region, output the pixel color value {R, G, B}, otherwise when in the blanking region, output {0, 0, 0}.
7. The output signals {*Hsync*, *Vsync*, vgaRed, vgaGreen, vgaBlue} should be defined as flip-flops to ensure no combinational logic delays will interfere with the output display.
8. Downsample the xcoord and ycoord by a factor of 8, brining the resolution to 100x75, which is easier to handle.
9. Don't forget to use a debouncer for the buttons.

Notes:

1. The 50Mhz pixel clock can be generated from the 100Mhz system clock.
2. Do not turn on user-LEDs in response to turning on the user-switches.
3. Perform a rigorous simulation to ensure precise timing, which must perfectly correspond to the timing diagrams in the theoretical background.

In the Lab:

Connect the Jtag cable from PC to FPGA board

Connect the VGA cable from the monitor to the FPGA board.

Program the FPGA with your implemented design and call an instructor to show him a correct functioning of your VGA interface.

Take a photo of a correct functioning of the monitor.

Submit:

In the report – submit the files:

VGA_Interface.v, VGA_Interface_tb.v, Drawer.v, VGA_Top.v, task2.xdc.

Provide a screenshot of the simulation and explain its principle signals behavior by pointing with an arrow to these signals' transitions and annotating them.

Explain which problems appeared during the lab and how did you overcome these problems. Attach a photo of the correct monitor operation.